A Variational Ensemble Transport Filter for Non-linear Data Assimilation

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Motivation

Data assimilation (DA) merges observations with a model of how a geophysical state evolves (Nychka and Anderson, 2010). Ensemble DA runs the model forward on n members, then *updates* the ensemble given new data, inferring the forecast distribution from those few samples. Standard methods such as the ensemble Kalman filter (Evensen, 1994; Katzfuss et al., 2016) make this tractable through restrictive assumptions.

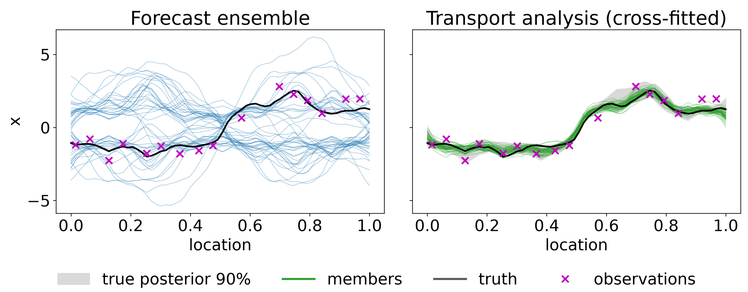


Figure 1. A bimodal forecast ensemble ($n = 50$, left) and our updated ensemble (right).

To avoid these assumptions, we learn the forecast distribution with a Bayesian transport map and update each member by variational inference, allowing non-Gaussian forecasts and accounting for uncertainty at small n.

Overview: Transport Each Ensemble Member

Ensemble DA alternates a *forecast*, which runs an expensive simulator forward from each member, and an *update*, which conditions the ensemble on new observations. We describe one update and drop the time index.

Setup. A latent field $\mathbf{x} = (x_1, \dots, x_N)^\top$, with $x_i = x(s_i)$, is observed with noise at a subset $\mathcal{S}$ of the locations:

$$p(\mathbf{y} \mid \mathbf{x}) = \prod_{i \in \mathcal{S}} p(y_i \mid x_i), \qquad \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)} \sim p(\mathbf{x}).$$

The goal is an updated ensemble $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(n)}$ from the posterior $p(\mathbf{x} \mid \mathbf{y}) \propto p(\mathbf{x})\, p(\mathbf{y} \mid \mathbf{x})$.

Two maps.

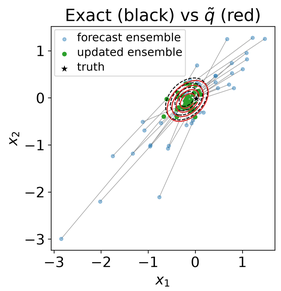
- $\mathcal{T}$: a Bayesian transport map (BTM) learned from the forecast ensemble, taking the learned forecast distribution to a standard normal, $\mathbf{z} = \mathcal{T}(\mathbf{x}) \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N)$.
- $\tilde{\mathcal{T}}$: a map learned by variational inference from $\mathbf{y}$, with $\tilde{\mathcal{T}}^{-1}(\mathbf{z}) \sim \tilde{q}(\mathbf{x}) \approx p(\mathbf{x} \mid \mathbf{y})$.

Update. Push every forecast member through both maps:

$$\tilde{\mathbf{x}}^{(j)} = \tilde{\mathcal{T}}^{-1}\big(\mathcal{T}(\mathbf{x}^{(j)})\big), \quad j = 1, \dots, n.$$

The image on the right shows the result of these transport steps. Each member keeps its latent position $\mathbf{z}^{(j)}$, so it moves from its place in the forecast (blue) to the matching place in the posterior (green).

In an operational DA setting, the updated ensemble would be simulated forward, and the update step would repeat with new observations at a future time point.



Two Sources of Error

At small n, ensemble updates are distorted by sampling error in two ways, which the EnKF usually patches with tuned variance inflation:

Problem	Our fix
The estimated forecast distribution is treated as known	The BTM integrates over its own estimation uncertainty (Step 1)
Inbreeding: each member is updated using an estimate computed from itself (Houtekamer and Mitchell, 1998)	Cross-fitting: each member is updated with maps learned without it (Step 3)

Step 1: Learn the Forecast Distribution

A Bayesian transport map (BTM) writes the forecast density as sparse nonlinear regressions with Gaussian process priors (Katzfuss and Schäfer, 2024):

$$p(\mathbf{x}) = \prod_{i=1}^{N} \mathcal{N}\big(x_i \mid f_i(\mathbf{x}_{c(i)}),\, d_i^2\big), \qquad f_i \sim \mathcal{GP},$$

where $c(i)$ indexes nearest previously ordered neighbors. Integrating out f_i and d_i gives a closed-form estimate $\hat{p}(\mathbf{x})$, a product of Student-t conditionals, and its map $\mathcal{T}$.

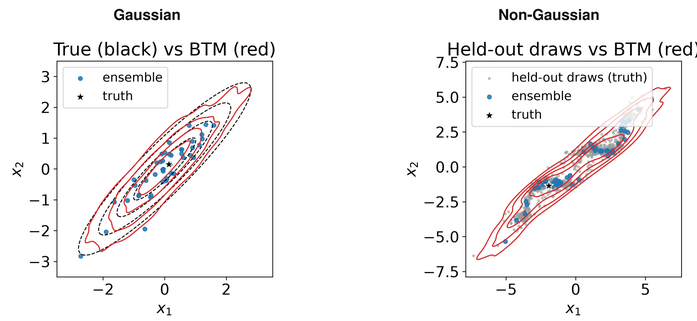


Figure 2. The BTM (red) learned from a small forecast ensemble (blue), for two locations.

Step 2: Update by Variational Inference

We approximate the posterior under the learned forecast distribution, $\hat{p}(\mathbf{x} \mid \mathbf{y}) \propto \hat{p}(\mathbf{x})\, p(\mathbf{y} \mid \mathbf{x})$, within a family $\mathcal{Q}$ that has the same sparsity as the BTM:

$$\tilde{q} = \operatorname*{arg\,min}_{q \in \mathcal{Q}} \mathrm{KL}\big(q(\mathbf{x}) \,\big\|\, \hat{p}(\mathbf{x} \mid \mathbf{y})\big), \qquad \tilde{\mathcal{T}}^{-1}(\mathbf{z}) \sim \tilde{q}.$$

The distribution that you can represent depends on how rich your variational family $\mathcal{Q}$ is. Here, we start with a family where each conditional of q is Gaussian with mean linear in its neighbors, but future work will generalize this to normalizing-flow conditionals.

Gaussian check. For a Gaussian forecast the exact posterior is known, and $\tilde{q}$ matches it.

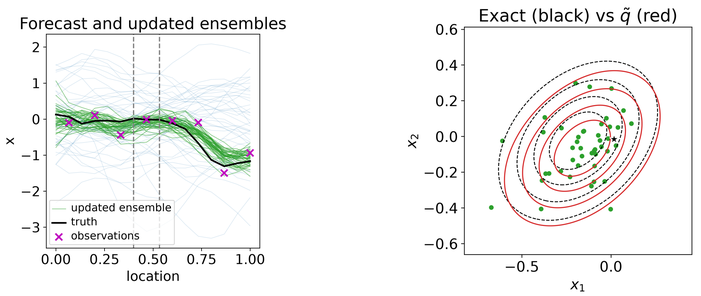


Figure 3. Left: forecast (blue) and updated (green) ensembles, truth (black), and observations (magenta); dashed lines mark the two locations shown on the right. Right: at these two locations, $\tilde{q}$ (red) matches the exact posterior (black).

Step 3: Cross-Fitting Prevents Inbreeding

Because $\mathcal{T}$ is learned from the members that we want to transport, it maps them too close to zero and the updated ensemble is too narrow. So, we instead update member j in fold k of K with maps learned without that fold:

$$\tilde{\mathbf{x}}^{(j)} = \big(\tilde{\mathcal{T}}^{(-k)}\big)^{-1}\big(\mathcal{T}^{(-k)}(\mathbf{x}^{(j)})\big).$$

For the BTM, the leave-one-out map has a closed form at the cost of the fit. For the variational update to the posterior, each fold requires its own fit.

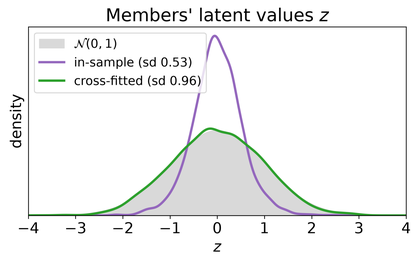


Figure 4. The members' latent values in one replicate: in-sample (purple) they are too concentrated; cross-fitted (green) they are close to standard normal.

Non-Gaussian Example

For non-Gaussian data, we can compare the updated ensembles for a standard EnKF update (with tapering) and our cross-fitted transport filter.

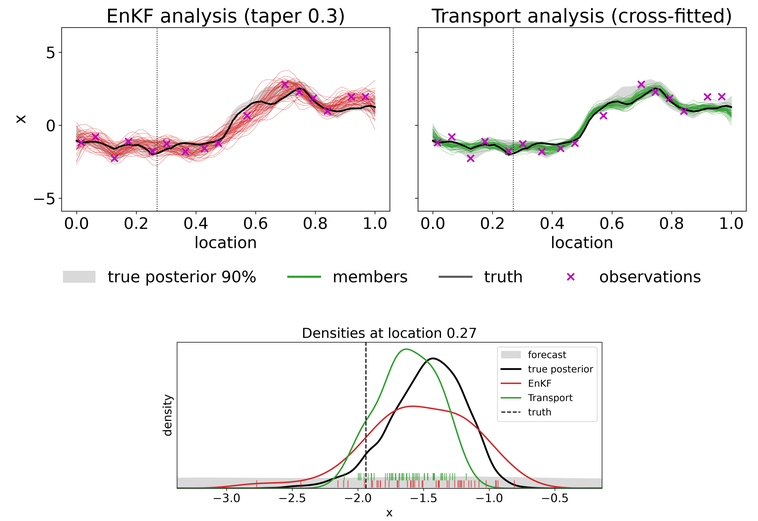


Figure 5. Top: EnKF and transport updates of the same forecast ensemble. Bottom: densities at the most bimodal unobserved location (dotted line above); ticks mark the individual members. The EnKF's linear update leaves the members too widely spread, while the transport update is much closer to the exact posterior.

Replicate Study

We repeated this experiment for 20 truths from the bimodal model ($N = 64$, $n = 50$, 16 observations, $\sigma_y = 0.5$, $K = 5$ folds) and scored each method against the truth. The table compares our updates with the forecast ensemble, a tapered EnKF, and RSIC-EnKF (Boyles and Katzfuss, 2021).

	Energy score	CRPS	90% coverage	Spread / exact
Forecast	9.038 (0.576)	0.947 (0.056)	0.91	7.62
EnKF (best taper)	2.170 (0.070)	0.219 (0.007)	0.89	1.76
RSIC-EnKF	1.951 (0.092)	0.196 (0.009)	0.89	1.55
Transport, naive	1.871 (0.106)	0.185 (0.010)	0.44	0.46
Transport, cross-fitted	1.687 (0.091)	0.165 (0.009)	0.94	0.94

Table 1. Means (standard errors) over replicates; fair energy score and CRPS (Gneiting et al., 2008; Ferro, 2014). Spread / exact: ensemble sd relative to the exact posterior's (1 is ideal).

Takeaways

1. The BTM learns Gaussian and non-Gaussian forecast distributions from small ensembles, accounting for its own estimation uncertainty.
2. In the Gaussian case, the variational update closely matches the exact update.
3. Cross-fitting removes inbreeding.
4. For a bimodal forecast, the transport update has a lower energy score than both the tapered EnKF and RSIC-EnKF in 15 of 20 replicates.
5. Future work will involve applying this to real-world data as well as extending the work to online parameter calibration via the filtering likelihood.

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